ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Phy-301 Classical Mechanics 2020

Solve the following: **(4x5=20)**

Q.2. If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange equation of motion, show by direct substitution that

$$L' = L + \frac{d}{dt}F(q_1, \cdots, q_n; t),$$

Q.3. Use Hamilton's principle of least action to derive Hamilton's equations of motion.

Q.4. The Lagrangian for two particles of masses m_1 and m_2 and coordinates x_1 and x_2, interacting via a potential $V(x_1 - x_2)$, is

$$L = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_1\dot{x}_2^2 - V(x_1 - x_2)$$

Rewrite the Lagrangian in terms of the center of mass coordinates

$$R = \frac{m_1\, x_1 + m_2\, x_2}{m_1 + m_2}$$

and the relative coordinates $x = x_1 - x_2$. Use Lagrange's equation to show that the center of mass and relative motion separate, the center of mass moving with constant velocity and relative motion being like that of a particle of reduced mass $\mu - \frac{m_1\ m_2}{m_1 + m_2}$ in a potentail $V(x)$.

Q.5. A pendulum of mass m is attached to a massless cart. The cart is attached with a spring of spring constant k and moves along x-axis. Determine the configuration space and find the Lagrange's equations of motion.

Solve the following. **(3x10=30)**

Q.6. (a) Assume Earth's orbit to be circular and that Sun's mass suddenly decreases by half. What orbit does Earth then have? Will Earth escape the solar system?

(b) Discuss the properties of motion in effective potential in a central force two-body problem. (5+5)

Q.7. (a) Use variational method to derive Euler-Lagrange equation of motion for a given functional

$$I = \int_{x_1}^{x_2} f\,[y(x), y'(x), x]dx.$$

(b) Show that the path followed by a particle in sliding from one point to another under the action of gravity in the absence of friction and in the shotest time is a cycloid. (5+5)

Q.8. (a) Show that the transformation

$$P = q\cot p,$$
$$Q = \log\left(\frac{\sin p}{q}\right),$$

is canonical.

(b) Define Poisson bracket of two dynamical variables. Show that the Poisson brackets are invariant under the canonical transformation. (5+5)

Q.1

Date:

Show $L' = L + \frac{d}{dt} F(q_1, q_2, \cdots, q_n, t)$

satisfies Lagrangian eq. of motion.

As, $\frac{d}{dt}\left(\frac{\partial L'}{\partial \dot{q}_i}\right) - \frac{\partial L'}{\partial q_i} = 0$

$\because L' = L + \frac{d}{dt} F(q_1, \cdots, q_n, t) = L + F'$

$$\frac{d}{dt}\left[\frac{\partial}{\partial \dot{q}_i}(L + F')\right] - \frac{\partial}{\partial q_i}(L + F') = 0$$

$$\left[\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i}\right] + \left[\frac{d}{dt}\left(\frac{\partial F'}{\partial \dot{q}_i}\right) - \frac{\partial F'}{\partial q_i}\right] = 0$$

where, $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0$

$$\frac{d}{dt}\left(\frac{\partial F'}{\partial \dot{q}_i}\right) - \frac{\partial F'}{\partial q_i} = 0$$

For $\frac{d}{dt}\left(\frac{\partial F'}{\partial \dot{q}_i}\right)$:-

$$F = F(q_1, \cdots, q_n, t)$$

$$\frac{dF}{dt} = \frac{\partial F}{\partial q_1}\frac{\partial q_1}{\partial t} + \frac{\partial F}{\partial q_2}\frac{\partial q_2}{\partial t} + \cdots + \frac{\partial F}{\partial q_n}\frac{\partial q_n}{\partial t} + \frac{\partial F}{\partial t}\frac{\partial t}{\partial t}$$

$$\frac{dF}{dt} = \frac{\partial F}{\partial q_1}\dot{q}_1 + \frac{\partial F}{\partial q_2}\dot{q}_2 + \cdots + \frac{\partial F}{\partial q_n}\dot{q}_n + \frac{\partial F}{\partial t}$$

$$F' = \sum_i \frac{\partial F}{\partial q_i}\dot{q}_i + \frac{\partial F}{\partial t}$$

$$\frac{\partial F'}{\partial \dot{q}_i} = \frac{\partial F}{\partial q_i} + 0$$

$$\frac{d}{dt}\left(\frac{\partial F'}{\partial \dot{q}_i}\right) = \frac{d}{dt}\left(\frac{\partial F}{\partial q_i}\right)$$

$$\frac{d}{dt}\left(\frac{\partial F'}{\partial \dot{q}_i}\right) = \frac{\partial \dot{F}}{\partial q_i} \quad \text{(Proved)}$$

$\therefore$ same as Lagrange eq. of motion.

Q.3

Derivation of Lagrange's equation from Hamiltonian Principle

$$J = \int_1^2 f(y, \dot{y}, x)$$

$$J = \int_1^2 f\left[y_1(x), y_2(x), \cdots, y_n(x), \dot{y}_1(x), \cdots, \dot{y}_n(x), x\right] dx$$

∴ all the quantities are the function of parameter 'x'.

$$J = \int_1^2 f\left[y_1(x,\alpha), y_2(x,\alpha), \cdots, y_n(x,\alpha), \dot{y}_1(x,\alpha), \cdots, \dot{y}_n(x,\alpha), x\right] dx$$

∴ all the quantities are the function of parameter 'α'.

$$y(x,\alpha) = y(x,0) + \alpha\eta(x)$$

$$y_1(x,\alpha) = y_1(x,0) + \alpha\eta_1(x)$$

$$\vdots$$

$$y_n(x,\alpha) = y_n(x,0) + \alpha\eta_n(x)$$

∴ $\eta_1, \eta_2 \cdots$ are completely arbitrary functions of x, which vanish at end points, i.e. at $x = x_1$ and $x = x_2$.

$$\frac{\partial J}{\partial \alpha} = \int_1^2 \left[\left(\frac{\partial f}{\partial y_1}\cdot\frac{\partial y_1}{\partial \alpha}\right) + \cdots + \left(\frac{\partial f}{\partial y_n}\cdot\frac{\partial y_n}{\partial \alpha}\right) + \left(\frac{\partial f}{\partial \dot{y}_1}\cdot\frac{\partial \dot{y}_1}{\partial \alpha}\right) + \cdots + \left(\frac{\partial f}{\partial \dot{y}_n}\cdot\frac{\partial \dot{y}_n}{\partial \alpha}\right) + \left(\frac{\partial f}{\partial x}\cdot\frac{\partial x}{\partial \alpha}\right)\right] dx$$

$$\frac{\partial J}{\partial \alpha} = \int_1^2 \sum_i \left[\left(\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha}\right) + \left(\frac{\partial f}{\partial \dot{y}_i}\cdot\frac{\partial \dot{y}_i}{\partial \alpha}\right)\right] dx$$

$\because \frac{\partial x}{\partial \alpha} = 0$

$$\frac{\partial J}{\partial \alpha} = \int_1^2 \sum_i \left[\left(\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha}\right) + \left(\frac{\partial f}{\partial \dot{y}_i}\cdot\frac{\partial}{\partial \alpha}\left(\frac{dy_i}{dx}\right)\right)\right] dx$$

$\because \dot{y}_i = \frac{dy_i}{dx}$

$$\frac{\partial J}{\partial \alpha} = \int_1^2 \sum_i \left[\left(\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha}\right) + \left(\frac{\partial f}{\partial \dot{y}_i}\cdot\frac{d}{dx}\left(\frac{\partial y_i}{\partial \alpha}\right)\right)\right] dx$$

$$\frac{\partial J}{\partial \alpha} = \int_1^2 \sum_i \left[\left(\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha}\right) + \left(\frac{\partial f}{\partial \dot{y}_i}\cdot\frac{\partial y_i}{\partial \alpha}\Big|_1^2 - \int_1^2 \frac{\partial y_i}{\partial \alpha}\cdot\frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i}\right)\right)\right] dx$$

∵ integration by parts

$$\frac{\partial J}{\partial \alpha} = \int_1^2 \sum_i \left[\left(\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha}\right) + \left(-\int_1^2 \frac{\partial y_i}{\partial \alpha}\cdot\frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i}\right) dx\right)\right]$$

∴ The term $\frac{\partial f}{\partial \dot{y}_i}\cdot\frac{\partial y_i}{\partial \alpha}\Big|_1^2$ vanishes because all curves passes through fixed end points.

$$\frac{\partial J}{\partial \alpha} = \int_1^2 \sum_i \left[\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha} - \frac{\partial y_i}{\partial \alpha}\cdot\frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i}\right)\right] dx$$

$$\left(\frac{\partial J}{\partial \alpha}\right)\cdot\delta\alpha = \int_1^2 \sum_i \left[\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha} - \frac{\partial y_i}{\partial \alpha}\cdot\frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i}\right)\right] \delta\alpha\, dx$$

$$\left(\frac{\partial J}{\partial \alpha}\right)_{\alpha=0} \delta\alpha = \int_1^2 \sum_i \left[\frac{\partial f}{\partial y_i}\cdot\frac{\partial y_i}{\partial \alpha} - \frac{\partial y_i}{\partial \alpha}\cdot\frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i}\right)\right]_{\alpha=0} \delta\alpha\, dx$$

∵ evaluating the integral at $\alpha = 0$.

$$\delta J = \int_1^2 \sum_i \left[\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i}\right)\right] \delta y_i$$

$\because \left(\frac{\partial J}{\partial \alpha}\right)_{\alpha=0} \delta\alpha = \delta J$

$\because \left(\frac{\partial y_i}{\partial \alpha}\right)_{\alpha=0} \delta\alpha = \delta y_i$

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for extremum condition, by using Hamiltonian principle, $\delta J = 0$:-

$$\int_1^2 \sum_i \left(\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i} \right) \right) \delta y_i \, dx = 0$$

from which :-

$$\sum_i \left(\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial \dot{y}_i} \right) \right) \delta y_i = 0$$

So,
$$\left(\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left[\frac{\partial f}{\partial \dot{y}_i} \right] \right] = 0 \quad \because \delta y_i \neq 0$$

$\because i = 1, 2, 3, \ldots, n$

Thus, the integral in Hamiltonian eq :-

$$I = \int_1^2 L(q_i, \dot{q}_i, t) \, dt$$

$$\because f(y_i, \dot{y}_i, x) \longrightarrow L(q_i, \dot{q}_i, t)$$

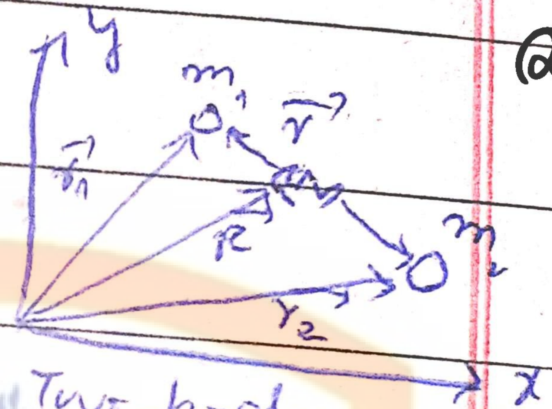
So,
$$\left(\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right] \right] = 0 \quad \because i = 1, 2, 3, \ldots, n$$

$\therefore$ shows that Langrange's eq. follow Hamiltonian's principle for conservative system.

... it's reduction to one-body problem and (reduced mass):-

$L = \frac{1}{2}m_1\dot{r}_1^2 + \frac{1}{2}m_2\dot{r}_2^2 - V(r)$ (Lagrangian of system)

Q.4



Two-body problem

For $\dot{r}_1$ and $\dot{r}_2$:

By definition of centre of mass:- $m_1\vec{r}_1{}' + m_2\vec{r}_2{}' = 0$

$\vec{r}_1{}' = -\frac{m_2}{m_1}\vec{r}_2{}'$

and $\vec{r}_2{}' = -\frac{m_1}{m_2}\vec{r}_1{}'$

Putting it in $\vec{r} = \vec{r}_1{}' - \vec{r}_2{}'$

$r = \vec{r}_1{}' + \frac{m_1}{m_2}\vec{r}_1{}' = \vec{r}_1{}'\left(1 + \frac{m_1}{m_2}\right)$

$\vec{r}_1{}' = \vec{r}\left(\frac{m_2}{m_1+m_2}\right)$

and $\vec{r}_2{}' = -\frac{m_1}{m_2}\left(\frac{m_2}{m_1+m_2}\right)\vec{r} = -\left(\frac{m_1}{m_1+m_2}\right)\vec{r}$

Now, $\vec{R} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2}$ (Position of centre of mass)

$\vec{R}(m_1+m_2) - m_2\vec{r}_2 = m_1\vec{r}_1$

$\because \vec{r} = \vec{r}_1 - \vec{r}_2$

~~$m_1\vec{R} + m_2\vec{R} - m_2\vec{r}_2 = m_1\vec{r}_1$~~

$m_1\vec{r_1} = (m_1+m_2)\vec{R} - m_2(\vec{r_1}-\vec{r})$

$m_1\vec{r_1} = (m_1+m_2)\vec{R} - m_2\vec{r_1} + m_2\vec{r}$

$(m_1+m_2)\vec{r_1} = (m_1+m_2)\vec{R} + m_2\vec{r}$

$\vec{r_1} = \vec{R} + \left(\frac{m_2}{m_1+m_2}\right)\vec{r}$

Thus, $\vec{r_2} = \vec{R} - \left(\frac{m_1}{m_1+m_2}\right)\vec{r}$

So, putting in eq 1 —

$L = \frac{1}{2}m_1\left(\vec{R} + \left(\frac{m_2}{m_1+m_2}\right)\dot{\vec{r}}\right)^2 +$

$\frac{1}{2}m_2\left(\vec{R} - \left(\frac{m_1}{m_1+m_2}\right)\dot{\vec{r}}\right)^2 - V(r)$

$\vec{L} = \frac{1}{2}m_1\left\{\vec{R}^2 + \left(\frac{m_2}{m_1+m_2}\right)^2\dot{r}^2 + 2\vec{R}\dot{\vec{r}}\left(\frac{m_2}{m_1+m_2}\right.\right.$

$\frac{1}{2}m_2\left\{\vec{R}^2 + \left(\frac{m_1}{m_1+m_2}\right)^2\dot{r}^2 - 2\vec{R}\dot{\vec{r}}\left(\frac{m_1}{m_1+m_2}\right.\right.$

$\vec{L} = \frac{1}{2}\vec{R}^2(m_1+m_2) + \frac{1}{2}\dot{r}^2\left(\frac{1}{m_1+m_2}\right)^2(m_1m_2)(m_1+$

$(m_1+m_2)\vec{R}\dot{\vec{r}}(m_1m_2)\left(\frac{1}{m_1+m_2}\right)^2 - V(r)$

$\vec{L} = \frac{1}{2}\vec{R}^2(m_1+m_2) - V(r) + \frac{1}{2}\dot{r}^2\left(\frac{1}{m_1+m_2}\right)(m_1m_2$

$\vec{R}\dot{\vec{r}}(m_2)\left(\frac{m_1}{m_1+m_2}\right) - m_2\left(\frac{m_1}{m_1+m_2}\right)\vec{R}\dot{\vec{r}}$

$\vec{L} = \frac{1}{2}MR^2 + \frac{1}{2}\dot{r}^2\mu - V(r)$

$\therefore \mu = \frac{1}{\frac{1}{m_1}+\frac{1}{m_2}}$

(reduced mass: the moving body behaves as if it's mass were the product of two masses divided by their sum)

Central forces

whose line of action passes through a single point/centre and the magnitude of it depends only on the distance from centre.

Find Lagrange's equations of motion for a pendulum attached to a massless rod further attached to a spring moving along x-axis.

Q.5

As, Lagrangian 'L':-

$$L = T - V$$

$$\therefore T = \frac{1}{2} M\dot{x}^2 + \frac{1}{2} m(\dot{x}_1^2 + \dot{y}_1^2)$$

where cartesian coordinates of the pendulum bob $(x_1, y_1) = (x + l\sin\theta, -l\cos\theta)$

So,
$$T = \frac{1}{2} M\dot{x}^2 + \frac{1}{2} m\left((\dot{x} + l\cos\theta\,\dot{\theta})^2 + l^2(\sin\theta\,\dot{\theta})^2\right)$$

$$\therefore V = \frac{1}{2} kx^2 + mg y_1$$

$$V = \frac{1}{2} kx^2 + mg(-l\cos\theta)$$

Putting:
$$L = \frac{1}{2} M\dot{x}^2 + \frac{1}{2} m(\dot{x}_1^2 + \dot{y}_1^2) - \frac{1}{2} kx^2 + mgl\cos\theta$$

Lagrange's equations are -

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0 \quad ; \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$$

Thus; $\frac{d}{dt}(\text{—}) - \text{—} = 0$; $\frac{d}{dt}(\text{—}) - \text{—} = 0$

$I = \int_{x_1}^{x_2} f[y(x), y'(x), x]\, dx$ will be min. only if it satisfies euler's eq. ?

So, $I[Y_\alpha] = \int_{x_1}^{x_2} f[y_\alpha(x), y'_\alpha(x), x]\, dx$

$$\therefore \left.\frac{dI}{d\alpha}\right|_{\alpha=0} = 0$$

Q.6(b)

Thus, $$\left.\frac{dI[Y_\alpha]}{d\alpha}\right|_{\alpha=0} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y_\alpha} \cdot \frac{\partial y_\alpha}{\partial \alpha} + \frac{\partial f}{\partial y'_\alpha} \cdot \frac{\partial y'_\alpha}{\partial \alpha} + \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \alpha}\right] dx$$

$\because Y_\alpha(x) = Y(x) + \underbrace{\alpha\,\eta(x)}_{\text{small variation}}$

$$\frac{\partial Y_\alpha(x)}{\partial \alpha} = \eta(x)$$

$$\left.\frac{dI[Y_\alpha]}{d\alpha}\right|_{\alpha=0} = \int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha}\eta(x) + \frac{\partial f}{\partial y'_\alpha}\eta'(x) + 0 \; dx$$

$$\int_{x_1}^{x_2} \left.\frac{dI[Y_\alpha]}{d\alpha}\right|_{\alpha=0} = \int_{x_1}^{x_2} \frac{\partial f}{\partial y_\alpha}\eta(x)\,dx + \left|\frac{\partial f}{\partial y'_\alpha}\eta(x)\right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx}\left(\frac{\partial f}{\partial y'_\alpha}\right)\eta(x)\,dx$$

$$\int_{x_1}^{x_2} \left.\frac{dI[Y_\alpha]}{d\alpha}\right|_{\alpha=0} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial Y_\alpha} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'_\alpha}\right)\right]\eta(x)\,dx$$

$$\therefore \frac{\partial f}{\partial Y_\alpha} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'_\alpha}\right) = 0 \quad \text{(Proved)}$$

$1 + e\cos\theta$

For a plane pendulum

Generalized coordinate

Constraint :- As length

conjineal/constraint

D'Alembert's principle

$\Sigma(F$

$\Sigma(F$

and $Q_r =$

$Q_\theta =$

The forces are

$T - mg\cos\theta$

Tension

$T = mL\dot{\theta}^2$

path followed by particle in sliding from one point to another in the shortest time is cycloid?

Q.7(a)

The Brachistochrone problem:-

concerned with minimizing the curve joining two points along which a particle fall under the action of gravity in shortest possible time.

By definition:-

$$v = dl/dt$$

$$\int dt = \int dl/v$$

$$\therefore \Delta T = \Delta V$$

(conservation of energy)

$$\tfrac{1}{2} m v^2 = mgx$$

$$v = \sqrt{2gx}$$

$$\therefore dl = \sqrt{1+y'^2}\,dx$$

(length element of curve)

$$t = \int \left(\sqrt{1+y'^2}\,dx\right)/\sqrt{2gx}$$

Thus, $f(x) = \dfrac{\sqrt{1+y'^2}}{\sqrt{2gx}}$

For t = minimum; differential w.r.t. y and y':-

$$\frac{\partial f}{\partial y} = 0 \; ; \; \frac{\partial f}{\partial y'} = \frac{1}{2\sqrt{1+y'^2}}(2y') \cdot \frac{1}{\sqrt{2gx}}$$

$$\frac{d}{dt}\left(\frac{\partial f}{\partial y'}\right) - \frac{\partial f}{\partial y} = 0$$

(Euler-Lagrange's eq.)

$$\frac{d}{dx}\left(\frac{y'}{\sqrt{2gx(1+y'^2)}}\right) - 0 = 0$$

$$\int d\left(\frac{y'}{\sqrt{2gx(1+y'^2)}}\right) = \int (0)\,dy$$

$$\frac{y'}{\sqrt{2gx(1+y'^2)}} = a$$

$$\frac{y'^2}{2gx(1+y'^2)} = a^2$$

$$\therefore g a^2 = 1/4k$$

$$\frac{y'^2}{x(1+y'^2)} = 1/4k$$

$$2ky'^2 = x(1+y'^2)$$

$$2ky'^2 = x + xy'^2$$

$$(2k - x)y'^2 = x$$

$$y' = \sqrt{\frac{x}{2k-x}}$$

$$\int dy = \int \sqrt{x/2k-x}\,dx$$

$$y = \int \sqrt{x/2k-x}\,dx$$

$$\therefore x = k(1-\cos\theta)$$

$$dx = k\sin\theta\,d\theta$$

(changing variables)

$$y = \int \sqrt{\frac{k(1-\cos\theta)}{2k - k(1-\cos\theta)}}\,k\sin\theta\,d\theta$$

$$y = \int \sqrt{\frac{(1-\cos\theta)}{(1+\cos\theta)}}\,k\sin\theta\,d\theta \cdot \frac{\sqrt{(1-\cos\theta)}}{\sqrt{(1-\cos\theta)}}$$

$$y = \int \sqrt{\frac{(1-\cos\theta)^2}{1-\cos^2\theta}}\,k\sin\theta\,d\theta$$

$$y = \int \frac{1-\cos\theta}{\sin\theta}\,k\sin\theta\,d\theta$$

y = ...

Date:

For E.L : $x = 0$

$$\boxed{x = k(1-\cos\theta)}$$

$$0 = k(1-\cos\theta)$$

$$1-\cos\theta = 0$$

$$\cos\theta = 1$$

$$\theta = 0$$

Putting it =

$$y = k(\theta - \sin\theta) + c$$

$$0 = k(0-0) + c$$

$$c = 0$$

So, $\boxed{y = k(\theta - \sin\theta)}$

← eq. of cycloid. For these values of x and y, time will be shortest.

are euler-eq^n of motion.

For a transformation to be canonical

$\sum (p\,dq_k - P\,dQ_k)$

For $P = q\cot p$; $Q = \log\left(\frac{\sin p}{q}\right)$

Thus; $p\,dq - P\,dQ$:-

$= p\,dq - q\cot p \frac{1}{1/q \sin p}\, d\left(\frac{\sin p}{q}\right)$

$= p\,dq - \cancel{q}\cot p \frac{1}{\sin p} \frac{q \cos p\,dp - \sin p\,dq}{q^{\cancel{2}}}$

$= p\,dq - \cot p \left(\frac{q\cos p\,dp}{\sin p} - dq\right)$

$= p\,dq - \cot p\,(q\cot p\,dp - dq)$

$= p\,dq - q\cot^2 p\,dp + \cot p\,dq$

Q.7(b)

$= (p + \cot p)\,dq - q\cot^2 p\,dp$

$= (p + \cot p)\,dq - q(\operatorname{cosec}^2 p - 1)\,dp$

$= (p + \cot p)\,dq + q(1 - \operatorname{cosec}^2 p)\,dp$

So; $= d((p + \cot p)\,q)$ $\therefore$ perfect differential (Proved)

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For the generating function $F(q, Q)$:-

$$dF = \frac{\partial F}{\partial q} dq + \frac{\partial F}{\partial Q} dQ$$

$$dF = +p\,dq - P\,dQ \qquad \because P, q.$$

So, $dF = \sin^{-1}(q e^{Q})\,dq - q\cot(\sin^{-1}(q e^{Q}))\,dQ$

$$P = q\cot p$$

and $Q = \log\left(\frac{\sin p}{q}\right)$

$$e^{Q} = \frac{\sin p}{q}$$

$$\sin^{-1}(q e^{Q}) = p$$

Putting it in P :-

$$P = q\cot(\sin^{-1}(q e^{Q}))$$

$$\frac{dF}{dq} = \sin^{-1}(q e^{Q}), \quad \frac{dF}{dQ} = q\cot(\sin^{-1}(q e^{Q}))$$

$F = \int$ ——— , $F = \int$ ———

Conical invariance of Poisson Brackets:-

As, Hamiltonian equations show conical invariance, so, Poisson brackets also show conical invariance :-

$$[u,v]_{q,p} = [u,v]_{Q,P}$$

if (q,p) and (Q,P) are two conical conjugate sets.

Q.8

Considering the first eq. of transformation as a function of q and Q :- $Q = Q(q,p)$ or $P = \phi(q,Q)$ $P = P(q,p)$

$u\ v$

$p\ \partial q$

Substituting it in second equation of Transformation :-

$$P = \psi(q, Q)$$

By these two conditions, transformation can be expected to be created by a generating function of first kind :-

$$P = \frac{\partial F_1(q, Q)}{\partial q} \qquad \therefore F_1(q, Q)$$

and, $$P = \frac{\partial F_1(q, Q)}{\partial Q}$$

It will become true, if above equations hold

$$\frac{\partial \phi}{\partial Q} = -\frac{\partial \psi}{\partial q}$$

Now, by definition, Poisson bracket of $[Q, P]$:-

$$[Q, P] = \frac{\partial Q}{\partial q}\frac{\partial P}{\partial p} - \frac{\partial Q}{\partial p}\frac{\partial P}{\partial q} = 1$$

where, $P = \psi(q, Q)$:-

$$P = \psi(q, Q)$$

$$\delta P = \frac{\partial \psi}{\partial q}\delta q + \frac{\partial \psi}{\partial Q}\delta Q$$

$$\delta P = \frac{\partial \psi}{\partial q}\delta q + \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\partial q}\delta q + \frac{\partial Q}{\partial \phi}\delta \phi\right]$$

$\therefore$ For $Q = \phi(q, Q)$, the first transformation condition is :-

$$Q = Q(q, \phi(q, Q))$$

$$\frac{\partial Q}{\partial Q} = \frac{\partial Q}{\partial p}\cdot\frac{\partial \phi}{\partial Q} = 1$$

So, $$\delta Q = \frac{\partial Q}{\partial q}\delta q + \frac{\partial Q}{\partial \phi}\delta \phi$$

$$\delta P = \frac{\partial \psi}{\partial q}\delta q + \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\partial q}\delta q + \frac{\partial Q}{\partial \phi}\delta \phi\right]$$

$$\frac{\delta P}{\delta q} = \frac{\partial \psi}{\partial q} + \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\partial q} + \frac{\partial Q}{\partial \phi}\frac{\partial \phi}{\delta q}\right] \qquad \because \frac{\delta p}{\delta q} = 0$$

$$\frac{\delta P}{\delta q} = \frac{\partial \psi}{\partial q} + \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\partial q} + \frac{\partial Q}{\partial \phi}\frac{\partial p}{\delta q}\right]$$

$$\frac{\delta P}{\delta q} = \frac{\partial \psi}{\partial q} + \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\partial q}\right]$$

Similarly :-

$$\frac{\delta P}{\delta p} = \frac{\partial \psi}{\partial q}\frac{\delta q}{\delta p} + \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\partial q}\frac{\delta q}{\delta p} + \frac{\partial Q}{\partial \phi}\frac{\partial \phi}{\delta p}\right]$$

$$\frac{\delta P}{\partial p} = \frac{\partial \psi}{\partial q}\frac{\delta q}{\delta p} + \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\partial q}\frac{\delta q}{\delta p} + \frac{\partial Q}{\delta p}\right] \qquad \because \frac{\delta p}{\delta p} = 1$$

$$\frac{\delta P}{\delta p} = \frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\delta p}\right]$$

Putting these values in the above eq :-

$$[Q \cdot P] = \frac{\partial Q}{\partial q}\frac{\partial \psi}{\partial Q}\left[\frac{\partial Q}{\delta p}\right] - \frac{\partial Q}{\partial p}\frac{\partial \psi}{\partial q} - \frac{\partial Q}{\partial p}\frac{\partial \psi}{\partial Q}$$

$$[Q \cdot P] = -\frac{\partial Q}{\partial p}\frac{\partial \psi}{\partial q} = 1$$

Also ; $$\frac{\partial Q}{\partial Q} = \frac{\partial Q}{\partial p}\cdot\frac{\partial \phi}{\partial Q} = 1$$

comparing:-
$$\frac{\partial \cancel{Q}}{\partial p} \cdot \frac{\partial \phi}{\partial \cancel{Q}} = -\frac{\partial \cancel{Q}}{\partial p} \frac{\partial \psi}{\partial q}$$

$$\boxed{\frac{\partial \phi}{\partial Q} = -\frac{\partial \phi}{\partial q}}$$

leads to the existence of generating function.